

Vinh Ly

Palo Alto, CA | vily437@outlook.com | (318) 202-8037
<https://www.linkedin.com/in/vinh-ly-14b56b50>

Summary

Software Engineer with **18 years** building large scale distributed systems across consumer video, marketplace, and search platforms. Owned end to end delivery of video ingest, live streaming, trust and safety, and ad serving pipelines using **C++**, **Python**, **Go**, **Java**, and **Spanner**. Shipped Live Premieres reaching top 10 streaming audiences on YouTube. Partnered with PM, ML, and policy teams at Google, Microsoft, and Whatnot. Current focus on payments infrastructure, recipient facing platforms, and operational tooling for cash transfer programs in low connectivity regions. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages: C++, Python, Java, Go, SQL, TypeScript, JavaScript, HTML, CSS, Ruby

Systems: Spanner, Bigtable, Kafka, gRPC, Kubernetes

Domains: video ingest, live streaming, trust and safety, marketplace integrity, payments

Engineering: distributed systems, API design, code review, on call

Tooling: CI/CD, data pipelines, A/B testing, system design, AWS

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present | Remote

- Owning end to end technical direction for live premieres features reaching top 10 concurrent viewer counts on the platform, covering backend services in **Java** and **Spanner** backed workflows.
- Mentored 5 senior engineers across partner teams and drove quarterly reliability reviews that cut on call pages by roughly 40% on owned services.
- Reduced live stream startup latency by approximately 30% through **gRPC** refactors, connection reuse, and kernel level tuning on **Linux** servers handling fanout.
- Designed replay and VOD pipelines integrating with **Bigtable** and **Kafka** to persist millions of live sessions for subsequent creator analytics dashboards.
- Partnered with policy and trust teams on moderation hooks reaching top 100ms p99 latency budgets across global **YouTube** traffic.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025 | Remote

- Partnered with trust and safety PMs to ship real time risk scoring that blocked top tier fraud rings within minutes of detection.
- Cut reviewer turnaround time by roughly 45% by rewriting triage tooling against **Spanner** with denormalized read paths and async **Kafka** signals.
- Owned marketplace integrity services in **Python** and **Go** that screened over 2 million listings monthly for fraud and policy violations.
- Built payments reconciliation jobs that processed \$100M+ in monthly GMV with idempotent **gRPC** contracts and audit friendly **Bigtable** schemas.
- On call for production services across two regions and authored runbooks that shortened MTTR by about 35%.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024 | Remote

- Managed a team of 6 engineers delivering live streaming ingest, transcoding, and DVR features on YouTube that supported top tier creator events and sports broadcasts.
- Owned trust and safety review pipelines that ingested user reports from 100+ countries and routed cases to specialist queues in **Go** and **Python**.
- Drove migration of critical services from monolithic stacks to **Spanner** with zero downtime rollouts, reducing schema change lead time by roughly 60%.
- Shipped **YouTube Live** HTML5 playback path that handled concurrent viewer counts in the millions across **Java** backend and **C++** media servers.
- Designed ad serving prototypes that improved click through rate tests by low double digits across early experimentation cohorts.

Software Engineer | Google | 02/2011 - 02/2015 | Remote

- Built search ranking components in **C++** and **Java** that served top tier query result assembly for billions of daily requests.
- Partnered with international teams in Zurich and Tokyo on timezone handoff coverage and follow the sun on call rotations.
- Owned reliability for 3 production clusters and led incident reviews that decreased repeat incidents by roughly 40%.
- Reduced index build time by around 25% via parallelization in **MapReduce** jobs and **Bigtable** sharding adjustments.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011 | Remote

- Improved data collection throughput by about 3x through kernel level ETW refactors and ring buffer redesign.
- Co authored distributed aggregation pipelines that fed **SQL Server** reporting stores used by executive dashboards.
- Mentored 2 interns and shipped a code review dashboard adopted across the 80 person division.
- Developed internal **C++** services for Windows platform telemetry reaching hundreds of millions of endpoints.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006 | Remote

- Authored unit tests and shipped the tool as part of the division wide build pipeline within the 12 week internship.
- Prototyped **COM** components in **C++** for an internal developer tools prototype used by 200+ engineers.

Education

B.S. Honors

Virginia Tech | 2004 - 2007